

Khawaja Yunus Ali University

School of Science and Engineering
Department of Computer Science and Engineering

Mid-term Examination: Summer – 2022

Program: B.Sc. in CSE (Batch – 13th) 1st Year 2nd Semester

Course Title: Physics-II

Course Code: PHY 1201

Time: 10 Minutes

Total Marks: 10 X 1 = 10

Student ID:	Batch:	Session:
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Part 1: MCQ

These are single best answer type of questions. Each of the following question or statement has five suggested answers or completions. Write down the most appropriate one on your answer script.

- What amount of energy is released in every fission reaction?
 - 13.6MeV
 - 17.6MeV
 - 135MeV
 - 200MeV
 - 931MeV
- For elastic collision, which is the correct statement between kinetic energy before (K_b) and after (K_a) collision?
 - $K_b > K_a$
 - $K_b < K_a$
 - $K_b > 0$ & $K_a = 0$
 - $K_b = 0$ & $K_a > 0$
 - $K_b = K_a$
- Which is related with thermos-electric thermometer?
 - Thermocouple
 - Bayle's law
 - Heat death of universe
 - Elastic collision
 - Entropy
- The resistance thermometer measures the temperature in the unit of
 - K
 - J
 - $^{\circ}\text{C}$
 - R
 - F
- Which one is the correct equation for wave number?
 - $K = R \left(\frac{1}{n_u} - \frac{1}{n_l} \right)$
 - $K = R \left(\frac{1}{n_l} - \frac{1}{n_u} \right)$
 - $K = \left(\frac{1}{n_u} - \frac{1}{n_l} \right)$
 - $K = R \left(\frac{1}{n_u^2} - \frac{1}{n_l^2} \right)$
 - $K = R \left(\frac{1}{n_l^2} - \frac{1}{n_u^2} \right)$

6. The temperature of the heat source of a Carnot's engine is 400K. It takes 840J heat energy from the source at this temperature and rejects heat energy of 630J to the cold sink. What is the temperature of the cold sink?
- 200K
 - 300K
 - 400K
 - 500K
 - 600K
7. For reversible process – (Where, P -Pressure, V -Volume, S -Entropy, T -temperature, W -external work done)
- $\Delta P = 0$
 - $\Delta V = 0$
 - $\Delta S = 0$
 - $\Delta T = 0$
 - $\Delta W = 0$
8. What is the correct relationship between R & K ?
- $K = R/N_a$
 - $R = K/N_a$
 - $K = R$
 - $R = KT$
 - $K = R \cdot N_a$
9. Which will be the lower orbit number for the Paschen series of a hydrogen atom?
- 1
 - 2
 - 3
 - 4
 - 5
10. What is value of the reduced Planks constant ($\hbar$)?
- $6.63 \times 10^{-34} Js$
 - $3.31 \times 10^{-34} Js$
 - $2.11 \times 10^{-34} Js$
 - $1.0557 \times 10^{-34} Js$
 - $0.53 \times 10^{-34} Js$

Invigilator Signature

Examiner Signature

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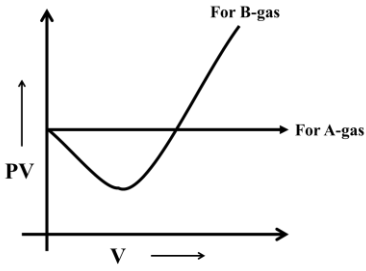
Time: 1 Hour and 20 Minutes

Total Marks: 20

Part 2: SAQ

(Answer Question No. 1 and any three from the rest.)

[The figures in the right margin indicate full marks for the respective question]

1.	See the stem and answer the question that follows:		
			
	a) Which graph satisfies the Boyles law properly?	[1]	CLO3
	b) At low temperature and high pressure, which conditions of A-gas are not true?	[1]	CLO1
	c) Construct the equation for A-gas by correcting the B-gas equation.	[3]	CLO2
2.	a) Define thermocouple.	[1]	CLO1
	b) Disclose the principle of temperature measurement.	[2]	CLO1
	c) A platinum resistance thermometer reads 6Ω and 8.5Ω at ice point and steam point respectively. Find the temperature at which it reads 12Ω.	[2]	CLO2
3.	a) What is degree of freedom in thermodynamics?	[1]	CLO1
	b) Is positive or negative Gibbs free energy favorable? Justify	[1]	CLO3
	c) Establish an argument by drawing a block diagram and coefficient of performance for "The refrigeration is the opposite process of heat engine".	[3]	CLO3 & CLO4
4.	a) State the properties of Carnot engine.	[1]	CLO1
	b) Is entropy continuously increasing? – give the mathematical logic.	[2]	CLO2 & CLO3
	c) Find the change in entropy in order to transform 5kg water of 10°C to vapour of 100°C?	[2]	CLO2
5.	See the short form of nuclear reaction and give answer of questions that follows:		
	${}^{26}_{12}\text{Mg}(p, \gamma)Y$ Where, mass of the proton = 1.00728amu, mass of the neutron = 1.00867amu, mass of the Mg nucleus = 24.3050amu and mass of the Y nucleus = 26.9815amu		
	a) Identify Y and write the full form the reaction.	[1]	CLO3
	b) Find the radius of the target nucleus.	[2]	CLO2
	c) Which is the most stable between target and product nucleus? - Justify your answer.	[2]	CLO4